SALES ANALYSIS AND SALES PREDICTION USING PYTHON AND POWER BI

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1. Abstract

This project focuses on analyzing sales data and predicting sales using Python and Machine Learning. Python libraries are used for data cleaning, exploratory data analysis, visualization, and building a regression model. Power BI is used to create an interactive dashboard to understand sales performance across countries, products, business segments, and years.

2. Objectives

. To clean and prepare sales data.

- To perform Exploratory Data Analysis (EDA).
- To identify sales and profit trends.
- To develop a regression model for sales prediction.
- To create an interactive Power BI dashboard.
- To provide useful business recommendations.

3. Tools and Technologies

- Python
- Pandas and NumPy
- Matplotlib and Seaborn
- Scikit-learn
- Power BI
- Microsoft Word

4. Dataset Description

The dataset contains 700 records and 16 original columns. It includes information about countries, products, business segments, units sold, prices, discounts, sales, profit, dates, and years.
During data cleaning, 53 missing values in the Discount Band column were handled by replacing them with “Unknown”. The Date column was converted into datetime format. No duplicate records were found.

5. Exploratory Data Analysis

EDA was performed to understand the dataset and identify important business patterns. Sales and profit were analyzed by country, product, business segment, and year. Bar charts were created to visualize the results.

Key Business Metrics

- **Total Sales:** 118,726,350.26
- **Total Profit:** 16,893,702.26
- **Total Units Sold:** 1,125,806

Major Findings

- Paseo generated the highest sales among the products.
- The Government segment contributed the highest sales.
- The United States of America recorded the highest sales among the listed countries.
- Sales and profit were considerably higher in 2014 than in 2013.

6. Sales Prediction Model

A Linear Regression model was developed using Scikit-learn to predict sales. The input features included segment, country, product, discount band, units sold,

manufacturing price, sale price, month, and year.

The dataset was divided into training and testing sets using an 80:20 split. The model was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score.

Model Results

- **MAE:** 82,876.54
- **RMSE:** 114,244.40
- **R^2 Score:** 0.7643

The R^2 score indicates that the model explains approximately 76.43% of the variation in sales on the test dataset. The model's predictions may help support sales planning, although prediction errors remain.

7. Power BI Dashboard

An interactive sales dashboard was developed in Power BI. It includes KPI cards for total sales, total profit, and total units sold, along with charts showing sales by country, product, and segment, and year-wise sales and profit.
The dashboard provides a clear overview of business performance and makes it easier to compare sales results across different categories.

8. Recommendations

- Focus on products with strong sales and profit performance.
- Analyze the reasons behind differences in sales across countries.
- Use year-wise trends to support future sales planning.

- Monitor business segments with lower sales and identify opportunities for improvement.
- Use the regression model as a supporting tool, not as a guarantee of future sales.

9. Conclusion

This project demonstrates how Python, Machine Learning, and Power BI can be used together for sales analysis and prediction. Data cleaning and EDA helped identify important business trends, while the Linear Regression model provided sales predictions with an R^2 score of 0.7643. The Power BI dashboard presents key findings in an accessible visual format. Overall, the project supports data-driven business analysis and planning.

10. Future Scope

Future improvements may include testing additional regression algorithms, using time-series forecasting, improving model accuracy, and adding more interactive filters and prediction features to the dashboard.